

Question:
Which type of articulators permit horizontal as well as vertical motion, but do not orient the motion to the TMJ via facebow transfer?
Answer Choices:
A. Class I
B. Class II
C. Class III
D. Class IV

Reasoning + Answer:

<tool_call>

To determine which type of articulator allows both horizontal and vertical motion without orienting motion to the TMJ via facebow transfer, we must understand the classification of dental articulators.

Class I articulators are designed to mimic the natural movement of the temporomandibular joint (TMJ). They use facebow transfer to orient the movement of the articulator to the TMJ, providing accurate simulation of jaw motion. Therefore, they do use facebow transfer and are not the correct answer.

Class II articulators allow for both horizontal and vertical movements. However, they are designed to orient the motion based on the facebow transfer, so they do use TMJ orientation via facebow transfer. Thus, they are not the correct choice.

Class III articulators are designed to simulate the TMJ motion more accurately and typically use facebow transfer to orient the movement. They are not the correct answer because they do orient motion via facebow transfer.

Class IV articulators are the most advanced and are designed to allow both horizontal and vertical movements. Crucially, Class IV articulators are designed to allow motion without requiring facebow transfer to orient the movement to the TMJ. This means the motion is not dependent on facebow transfer for orientation, and thus the articulator can simulate TMJ motion without the need for facebow transfer.

Therefore, the articulator that permits both horizontal and vertical motion but does not orient the motion to the TMJ via facebow transfer is the Class IV articulator.

<answer>

D. Class IV

</answer>